

PRACTICAL-05

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

CODE:-

```
venu@LAPTOP-KHLR30TF: ~/os-lab/practical$ cat practical-05.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>
#include <time.h>

int main()
{
    int pipefd[2];
    pid_t pid;

    char buffer[100];
    char message[] = "Hello from Producer!";

    struct timespec start, end;
    double elapsed;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    clock_gettime(CLOCK_MONOTONIC, &start);

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid > 0)
    {
        // Parent = Producer
        close(pipefd[0]);

        printf("Producer (Parent)\n");

        // Parent = Producer
        close(pipefd[0]);

        printf("Producer (Parent)\n");
        printf("Parent PID: %d\n", getpid());
        printf("Sending: %s\n", message);

        write(pipefd[1], message, strlen(message) + 1);

        close(pipefd[1]);

        wait(NULL);

        clock_gettime(CLOCK_MONOTONIC, &end);

        elapsed = (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1000000000.0;

        printf("Communication completed.\n");
        printf("Communication time: %.6f seconds\n", elapsed);
    }
    else
    {
        // Child = Consumer
        close(pipefd[1]);

        read(pipefd[0], buffer, sizeof(buffer));

        printf("\nConsumer (Child)\n");
        printf("Child PID: %d\n", getpid());
        printf("Received: %s\n", buffer);

        close(pipefd[0]);

        exit(0);
    }

    return 0;
}
venu@LAPTOP-KHLR30TF:~/os-lab/practical$
```

OUTPUT:-

```
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ nano practical-05.c
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ gcc practical-05.c -o practical-05
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ^C
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ./practical-05
Producer (Parent)
Parent PID: 1074
Sending: Hello from Producer!

Consumer (Child)
Child PID: 1075
Received: Hello from Producer!
Communication completed.
Communication time: 0.000685 seconds
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ cat practical-05.c
```